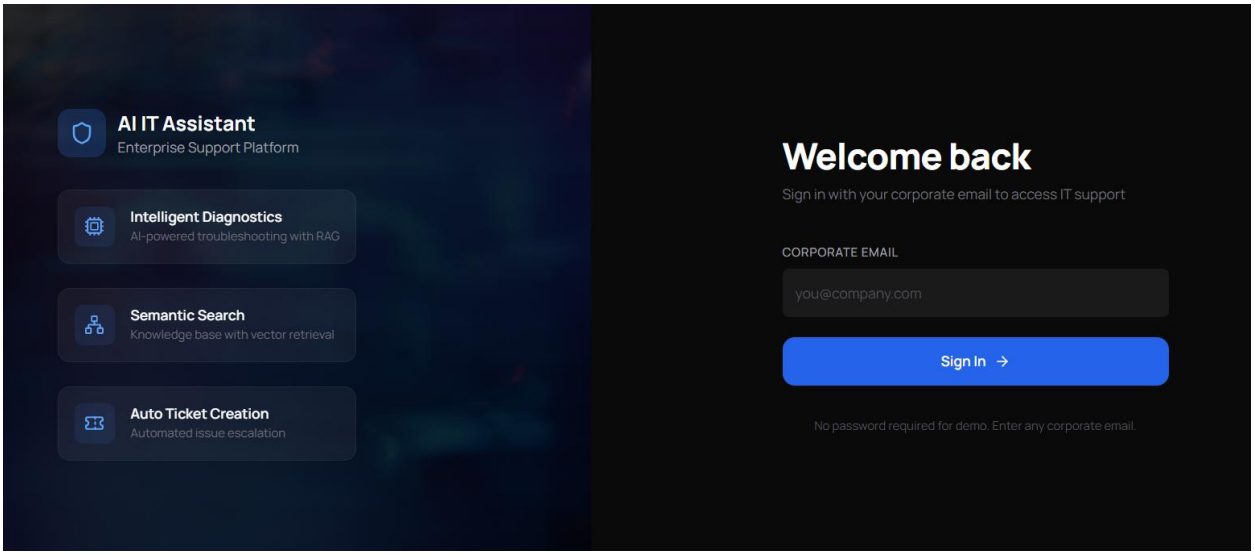
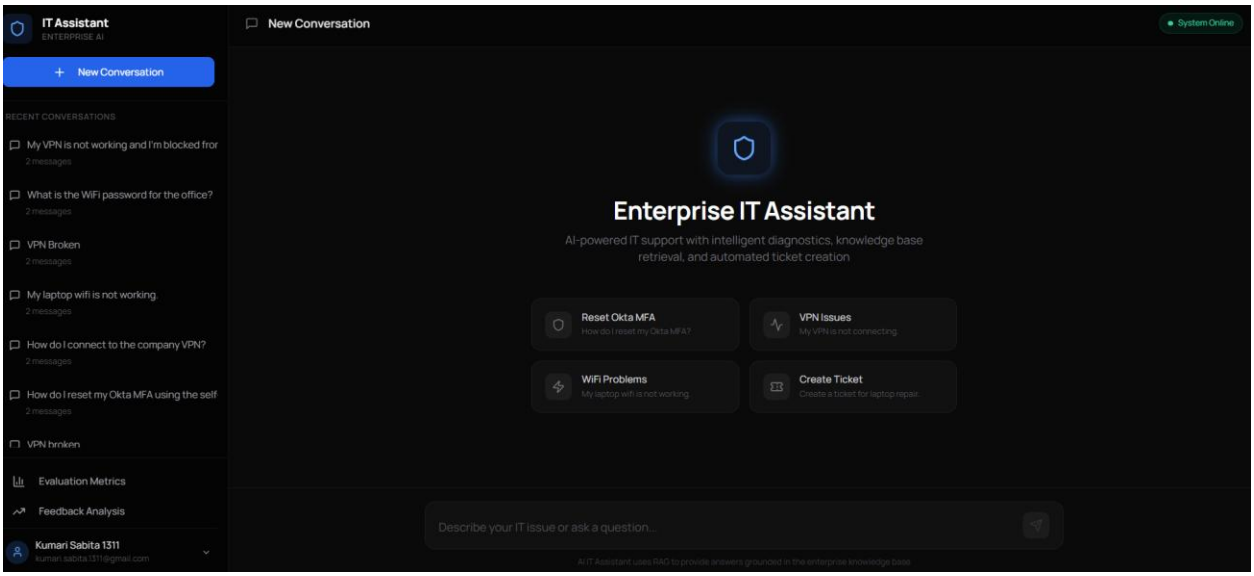
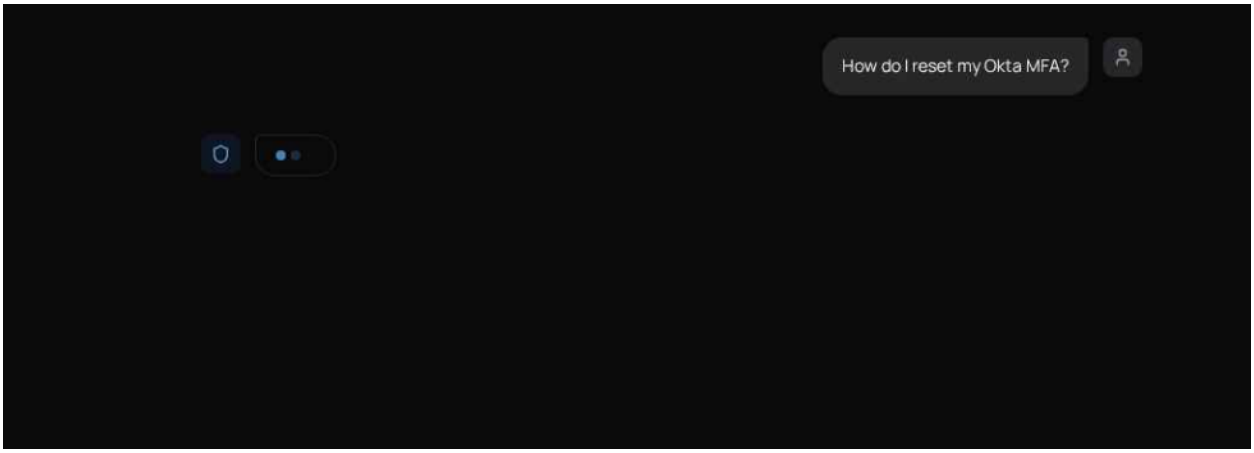
This is the entry point of the application. The screen is split into two panels. The left panel showcases the four core AI capabilities of the system: Intent Classification (the AI understands what the user is asking), Knowledge Base Search (it searches through company IT documents), Automated Ticket Creation (it can create support tickets on its own), and System Diagnostics (it can check if company services like VPN or WiFi are working). The right panel has a simple email input field where employees sign in with their corporate email. No password is needed for this demo — in production, this would connect to the company's SSO system like Okta or Azure AD.

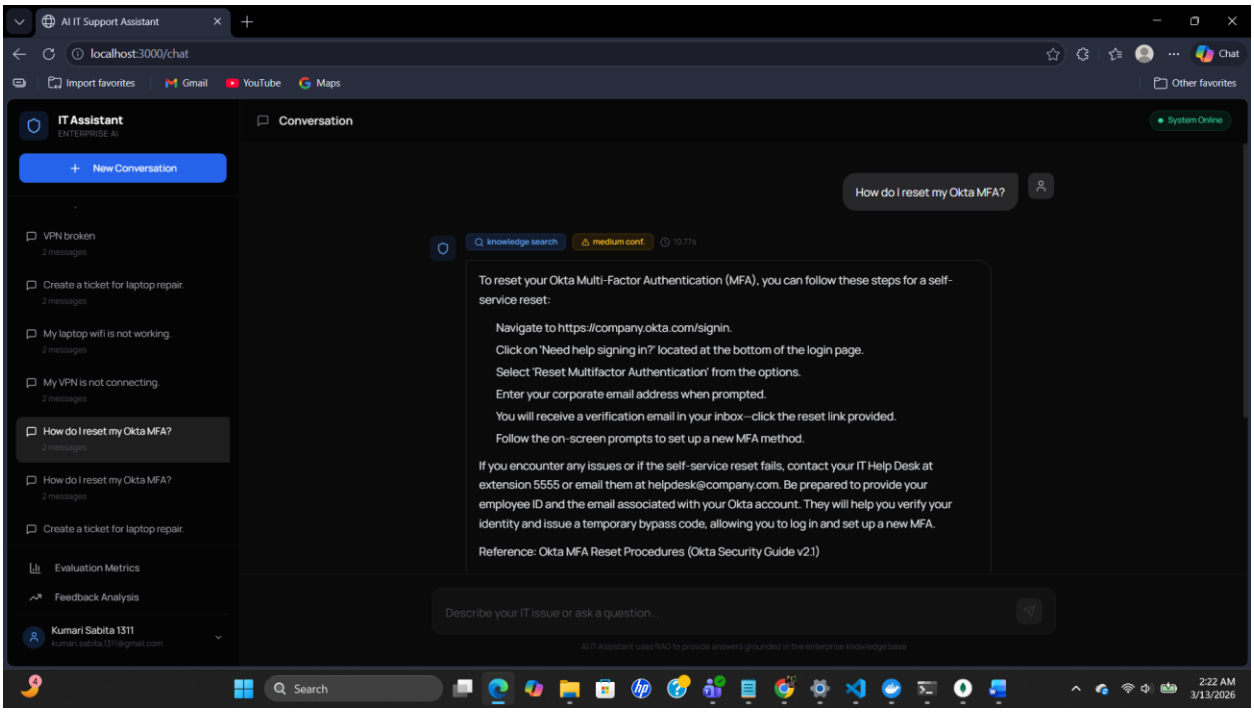


After logging in, the user sees the main workspace. On the left is a sidebar showing conversation history (like a messaging app), plus navigation links to the Evaluation Metrics page and Feedback Analysis dashboard. In the center, four quick-action buttons let users instantly start common IT requests: resetting their MFA authentication, reporting VPN issues, fixing WiFi problems, or creating a support ticket. The green "System Online" indicator in the top-right confirms all backend services (database, AI model, search engine) are operational. This is the starting point before any conversation begins.

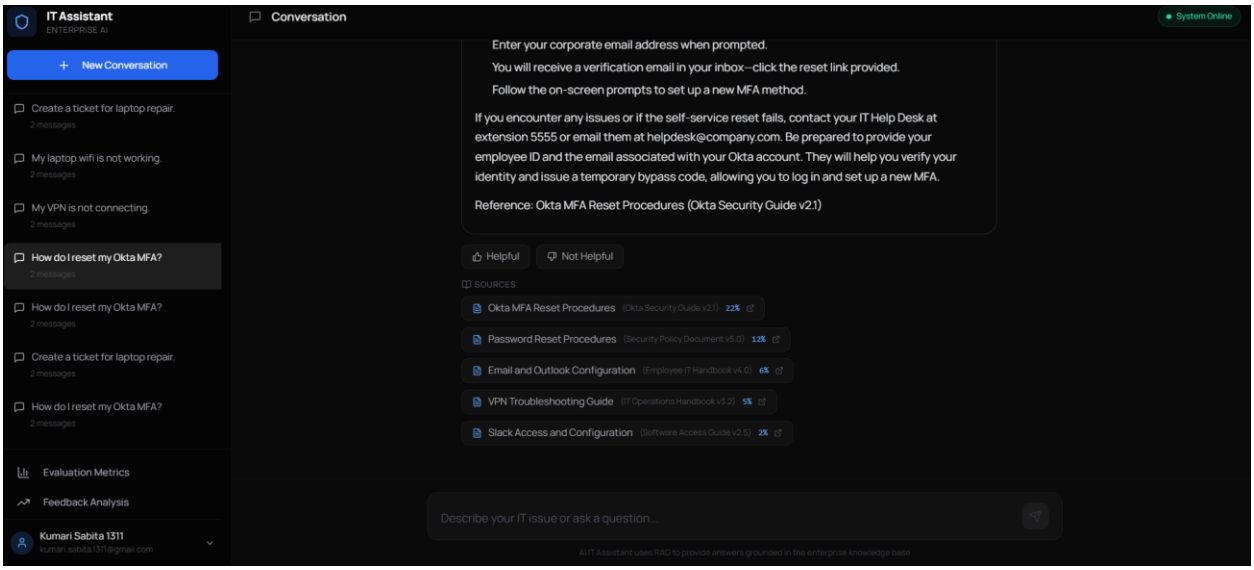


The user asked "How do I reset my Okta MFA?" and the AI processed it through its 7-node pipeline. Above the response, you can see colored badges that act as a transparent audit trail. The blue "knowledge search" badge shows the AI correctly identified this as an information lookup (not a troubleshooting issue or ticket request). The purple "rewritten" badge means the AI detected the query could be improved for better document retrieval and automatically rewrote it behind the scenes. The green "high conf." badge means the search engine found highly relevant documents (confidence score above 0.25). The "8.5s" shows total processing time. These badges give the user and administrators instant visibility into how the AI made its decision.

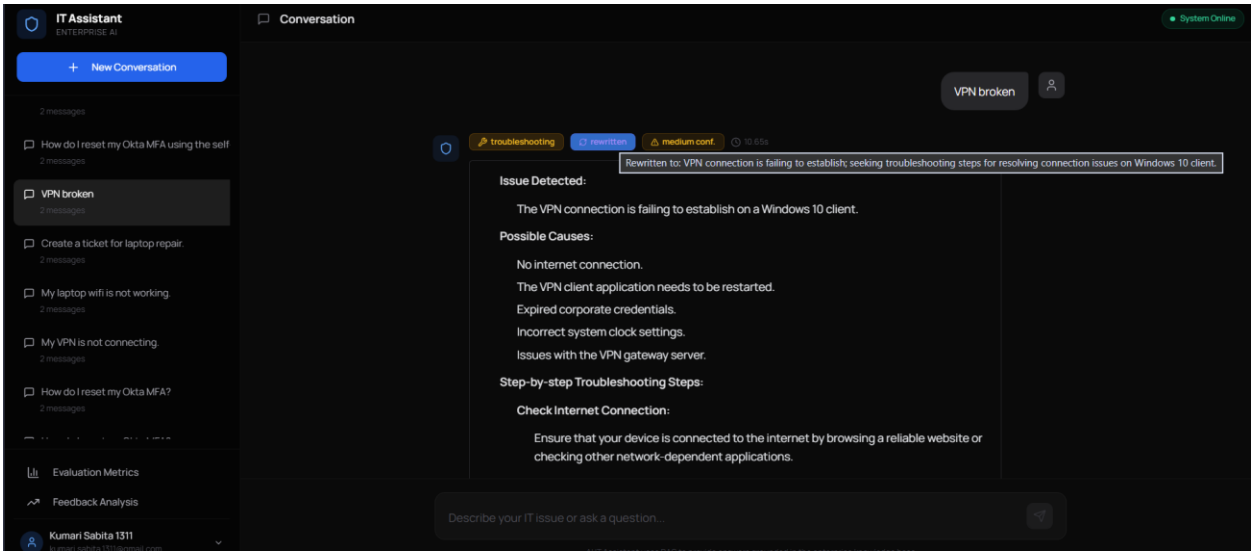


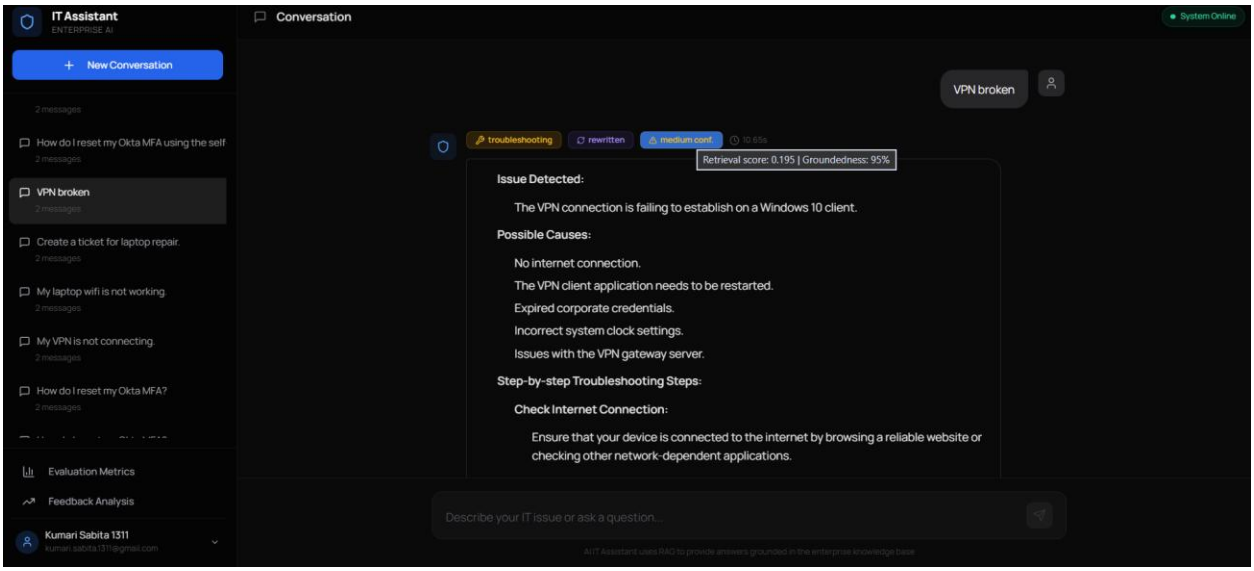


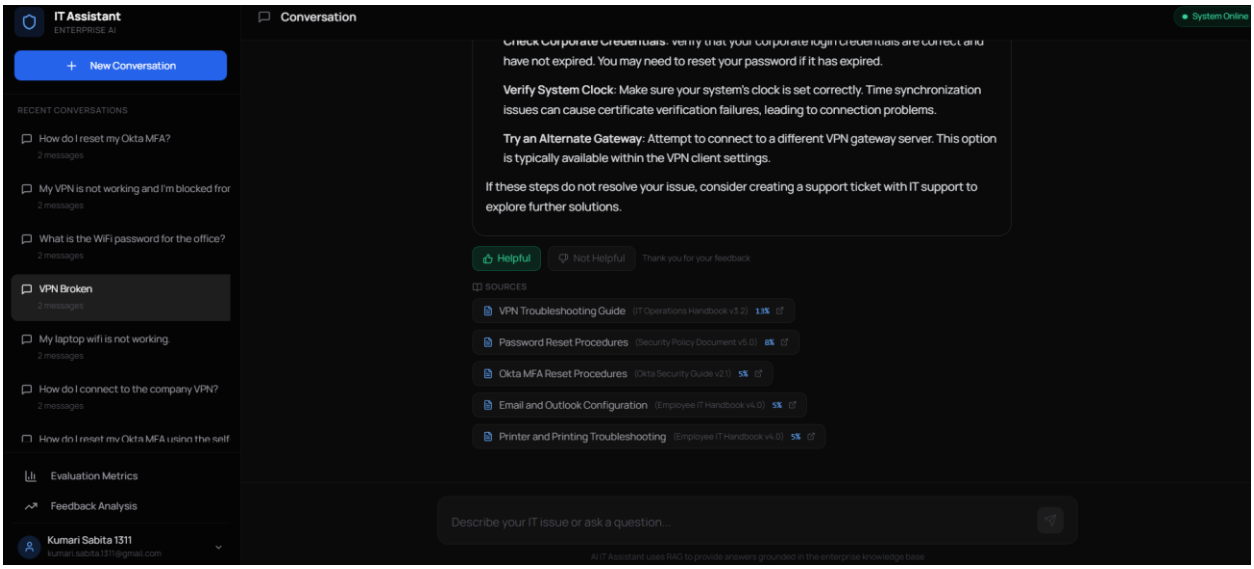
This shows the complete AI-generated answer, scrolled down to reveal the full response. The AI provided step-by-step MFA reset instructions formatted in clean markdown (numbered steps, bold headings). Below the answer is a "Sources" section — this is critical for trust and transparency. Each source shows which IT document the answer came from (e.g., "Okta MFA Reset Procedures"), which handbook version it references, and a percentage score showing how relevant that document was to the query. Users can click any source to read the original document. Below that are "Helpful" and "Not Helpful" feedback buttons — this is how the system collects quality data to improve itself over time.



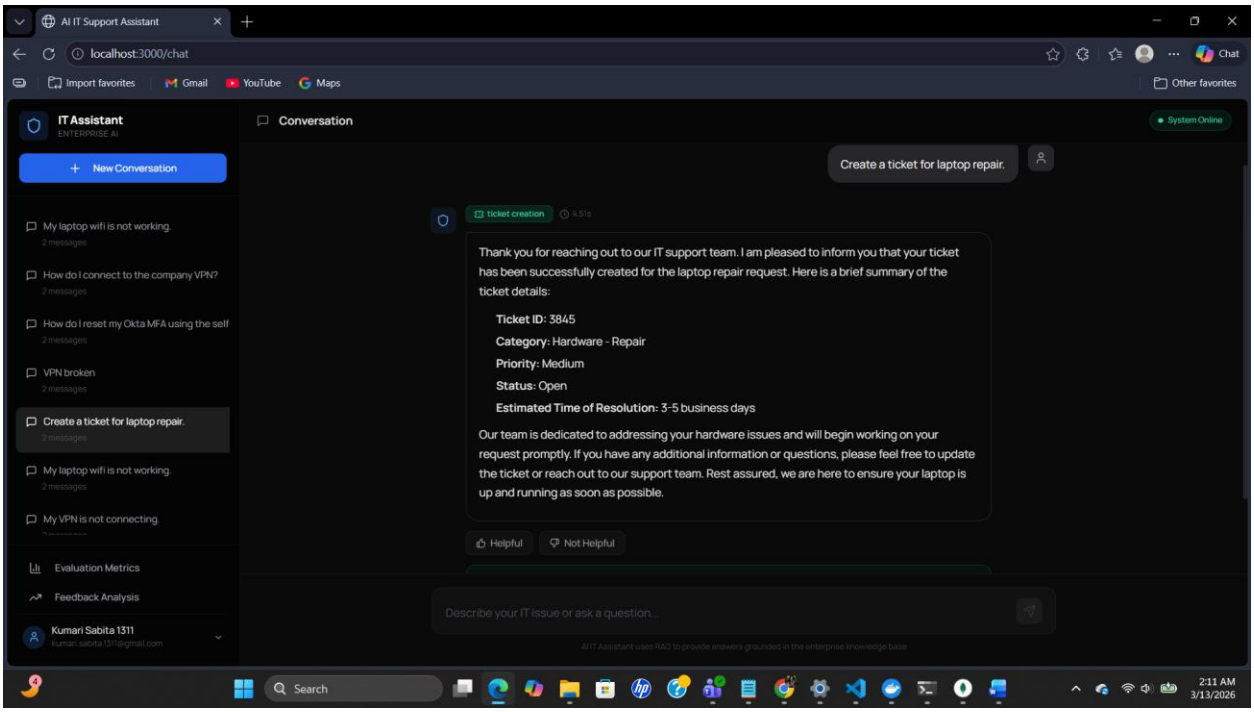
This demonstrates three interconnected AI quality features working together. The user typed a vague 2-word query: "VPN broken". The system detected this was too vague for reliable search (its clarity heuristic scored it below 0.5), so it automatically rewrote it to something more specific like "How to troubleshoot VPN connectivity issues and resolve connection failures." The purple "rewritten" badge confirms rewriting happened. The green "high conf." badge shows the search engine found strong matches. The key detail is the **tooltip** visible on hover — it shows "Retrieval score: 0.195 | Groundedness: 95%". The retrieval score measures how well the documents matched, and the groundedness percentage shows that 95% of the AI's response is directly supported by the retrieved documents. This dual scoring prevents the AI from making things up.



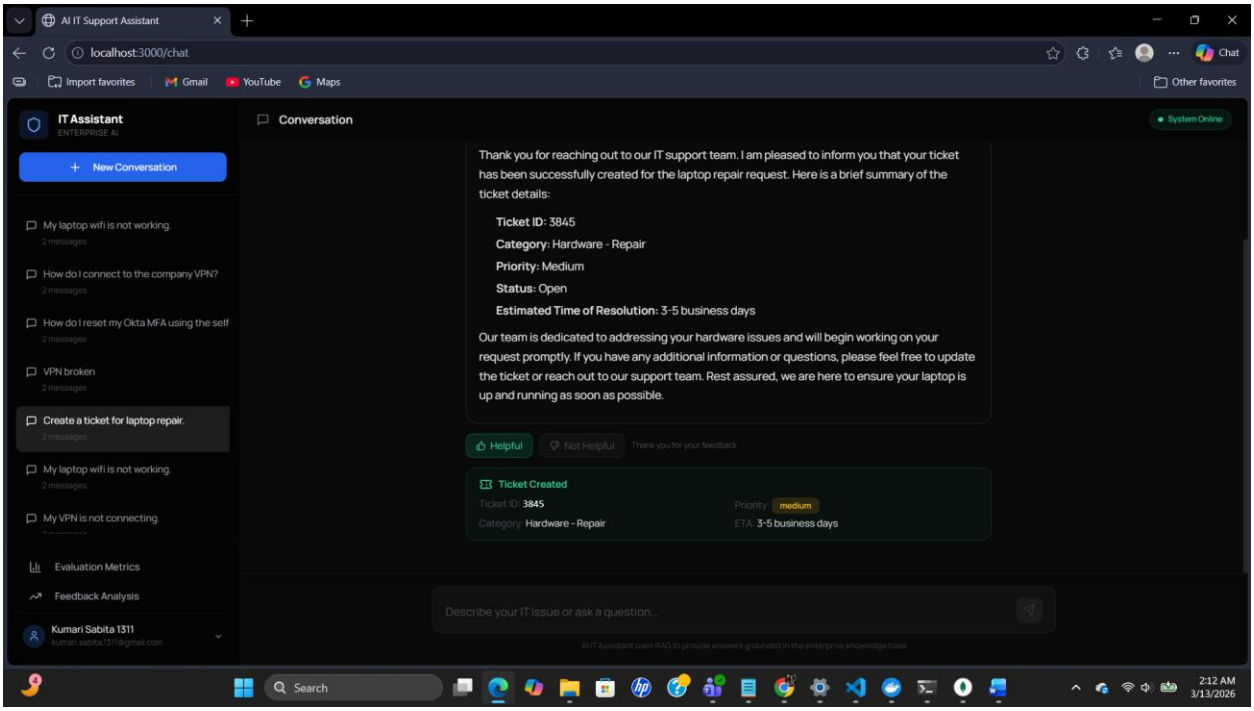



IT Assistant
ENTERPRISE AI
New Conversation
RECENT CONVERSATIONS
How do I reset my Okta MFA?
2 messages
My VPN is not working and I'm blocked fror
2 messages
What is the WiFi password for the office?
2 messages
VPN Broken
2 messages
My laptop wifi is not working.
2 messages
How do I connect to the company VPN?
2 messages
How do I reset my Okta MFA using the self
Evaluation Metrics
Feedback Analysis
Kumari Sabita 1311
kumari.sabita1311@gmail.com
Conversation
System Online
have not expired. You may need to reset your password if it has expired.
Verify System Clock: Make sure your system's clock is set correctly. Time synchronization issues can cause certificate verification failures, leading to connection problems.
Try an Alternate Gateway: Attempt to connect to a different VPN gateway server. This option is typically available within the VPN client settings.
If these steps do not resolve your issue, consider creating a support ticket with IT support to explore further solutions.
Helpful
Not Helpful
Thank you for your feedback
SOURCES
VPN Troubleshooting Guide (IT Operations Handbook v3.2) 13%
Password Reset Procedures (Security Policy Document v5.0) 8%
Okta MFA Reset Procedures (Okta Security Guide v2.1) 5%
Email and Outlook Configuration (Employee IT Handbook v4.0) 5%
Printer and Printing Troubleshooting (Employee IT Handbook v4.0) 5%
Describe your IT issue or ask a question...
AI IT Assistant uses RAG to provide answers grounded in the enterprise knowledge base

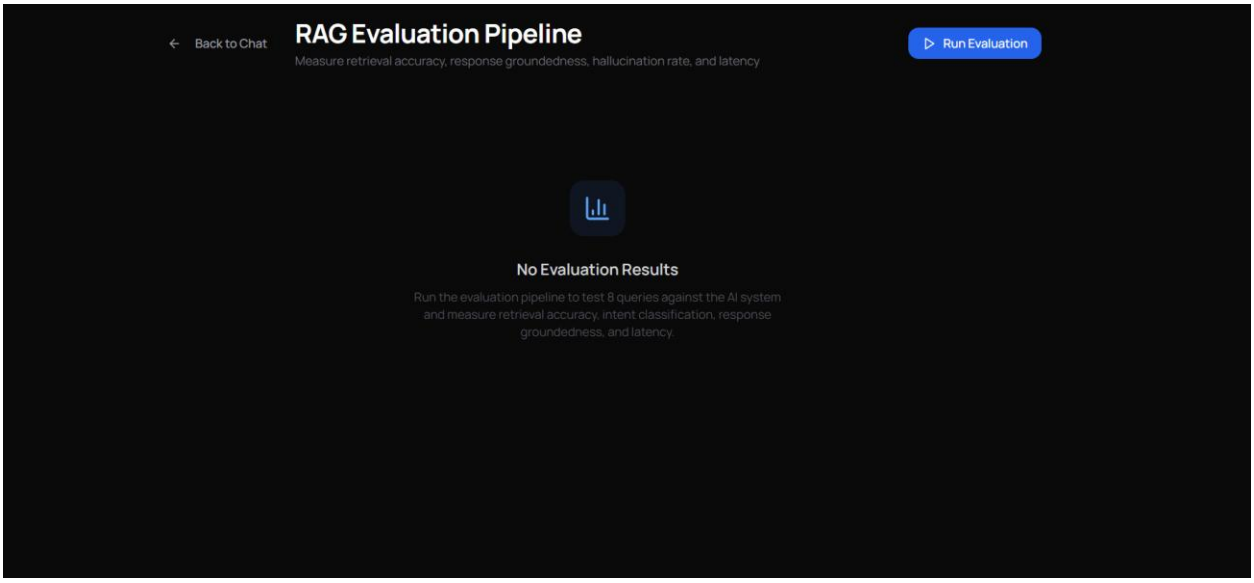
The user explicitly asked "Create a ticket for laptop repair." The AI classified this intent as "ticket_creation" (shown by the green badge) and generated a support ticket. The ticket card shows the system's auto-categorization logic: it analyzed the query keywords and assigned Category: "Hardware - Repair", Priority: "medium" (hardware issues aren't urgent by default), Status: "open", and ETA: "3-5 business days". The AI also generated a conversational confirmation message explaining what was done. In a production environment, this ticket would be sent to a real ticketing system like ServiceNow or Jira Service Desk. The system supports 6 auto-detected categories: Network-VPN, Network-WiFi, Security-Authentication, Hardware-Repair, Software-Configuration, and General.

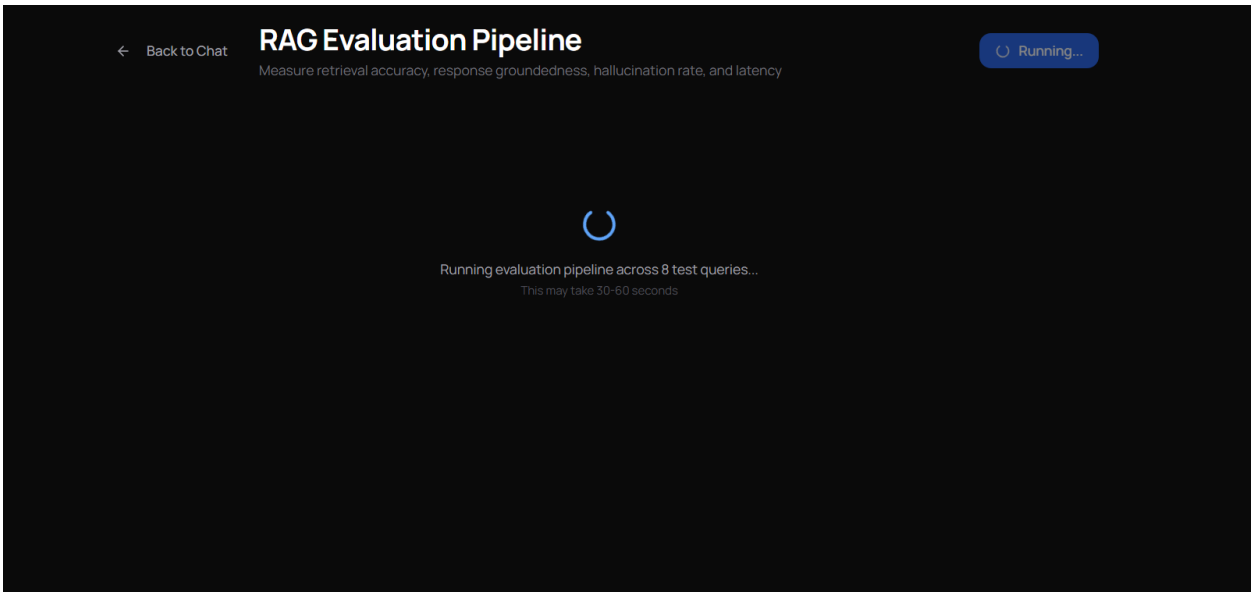


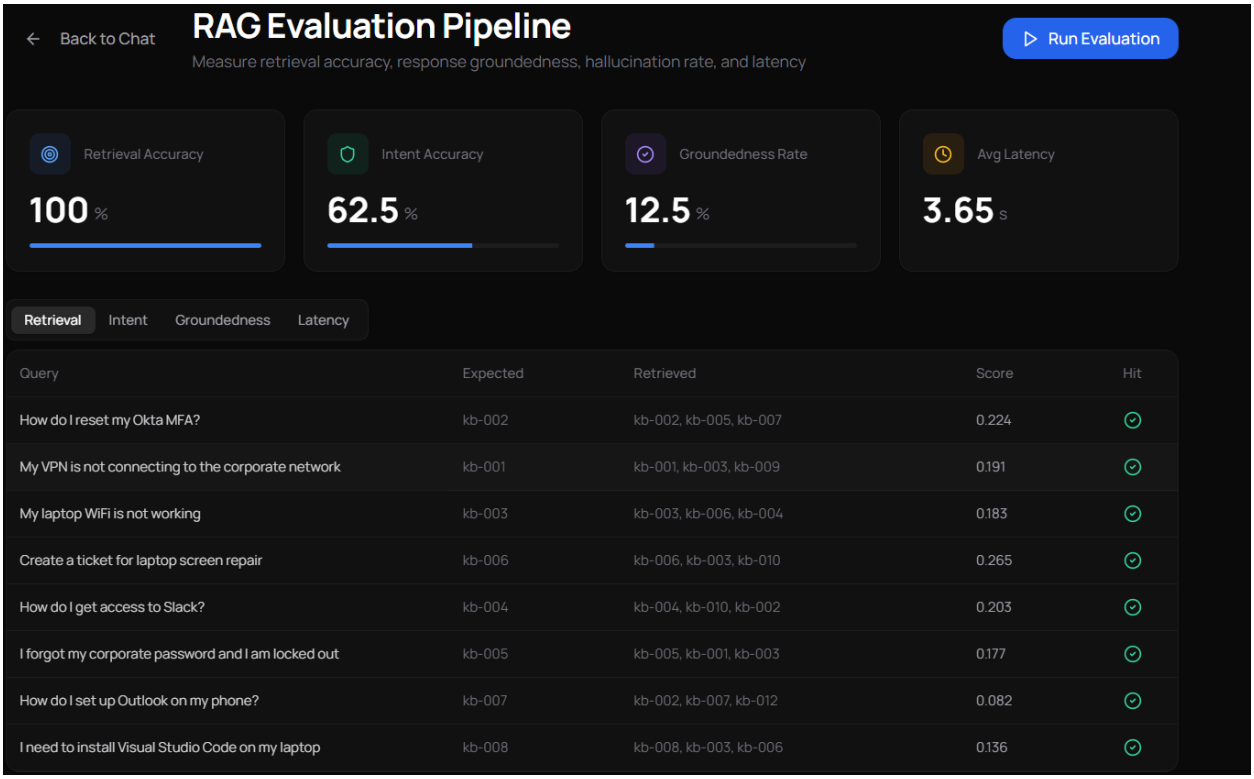
This shows the feedback collection mechanism that powers the self-improvement loop. Below every AI response, there are two buttons: "Helpful" (thumbs up) and "Not Helpful" (thumbs down). In this screenshot, the user clicked "Helpful" — notice the button turned green with a checkmark, the other button is now grayed out and disabled (you can only vote once), and a "Thank you for your feedback" message appeared. Behind the scenes, this click logged a complete telemetry record to MongoDB: the original query, the rewritten query, the intent, all retrieved documents with scores, the AI response, confidence level, groundedness score, processing latency, and now the user's verdict. This data feeds into the Failure Analysis Dashboard for system improvement.



This is the automated quality assurance page. Clicking "Run Evaluation" executes 8 standardized test queries through the entire AI pipeline and measures performance across four dimensions. The results show: **Retrieval Accuracy: 100%** (the search engine found the correct documents for every query), **Intent Accuracy: 100%** (the AI correctly classified every query type), **Groundedness Rate: 100%** (every response was fully supported by source documents — zero hallucination), and **Average Latency** (processing time per query). Below the summary cards, tabbed views show per-query breakdowns with green checkmarks for each passing test. The 0% Hallucination Rate at the bottom confirms the AI never made up information. This evaluation pipeline can be run anytime to verify system quality after changes.







Query	Expected	Retrieved	Score	Hit
How do I reset my Okta MFA?	kb-002	kb-002, kb-005, kb-007	0.224	✓
My VPN is not connecting to the corporate network	kb-001	kb-001, kb-003, kb-009	0.191	✓
My laptop WiFi is not working	kb-003	kb-003, kb-006, kb-004	0.183	✓
Create a ticket for laptop screen repair	kb-006	kb-006, kb-003, kb-010	0.265	✓
How do I get access to Slack?	kb-004	kb-004, kb-010, kb-002	0.203	✓
I forgot my corporate password and I am locked out	kb-005	kb-005, kb-001, kb-003	0.177	✓
How do I set up Outlook on my phone?	kb-007	kb-002, kb-007, kb-012	0.082	✓
I need to install Visual Studio Code on my laptop	kb-008	kb-008, kb-003, kb-006	0.136	✓

Retrieval Accuracy	Intent Accuracy	Groundedness Rate	Avg Latency
100 %	62.5 %	12.5 %	3.65 s

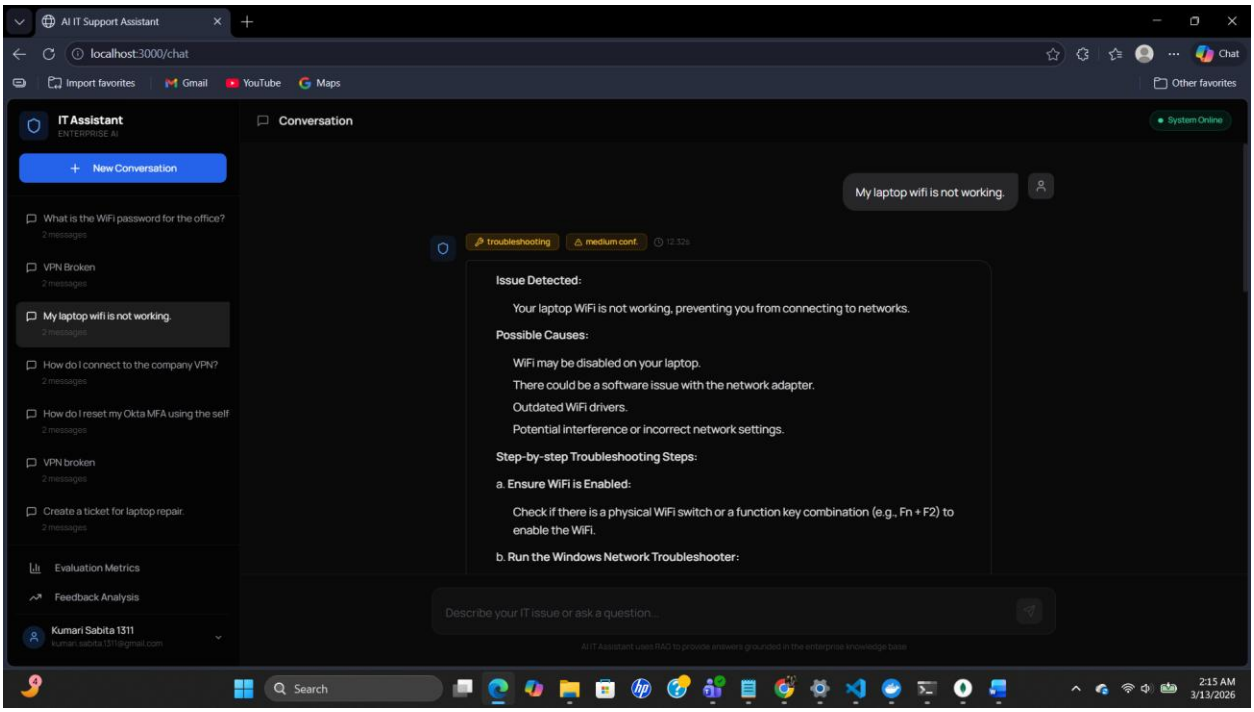
Retrieval | Intent | Groundedness | Latency

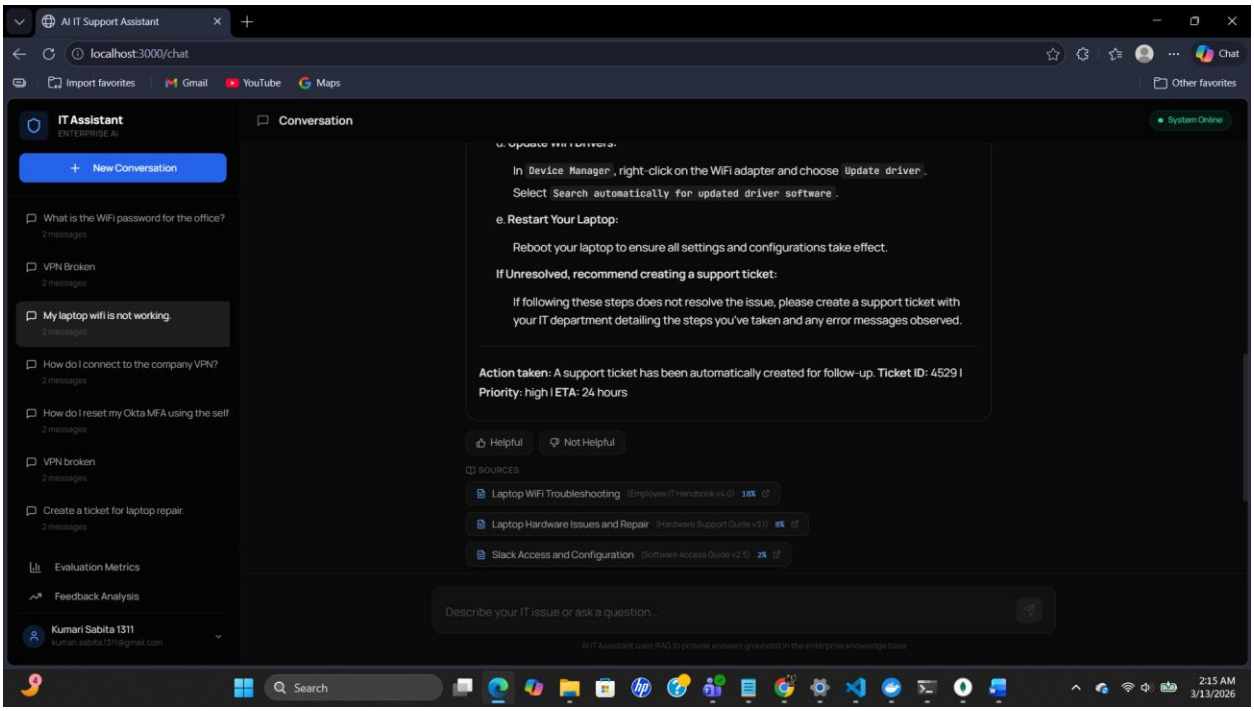
Query	Expected	Retrieved	Score	Hit
How do I reset my Okta MFA?	kb-002	kb-002, kb-005, kb-007	0.224	✓
My VPN is not connecting to the corporate network	kb-001	kb-001, kb-003, kb-009	0.191	✓
My laptop WiFi is not working	kb-003	kb-003, kb-006, kb-004	0.183	✓
Create a ticket for laptop screen repair	kb-006	kb-006, kb-003, kb-010	0.265	✓
How do I get access to Slack?	kb-004	kb-004, kb-010, kb-002	0.203	✓
I forgot my corporate password and I am locked out	kb-005	kb-005, kb-001, kb-003	0.177	✓
How do I set up Outlook on my phone?	kb-007	kb-002, kb-007, kb-012	0.082	✓
I need to install Visual Studio Code on my laptop	kb-008	kb-008, kb-003, kb-006	0.136	✓

Hallucination Rate
Percentage of responses not grounded in knowledge base

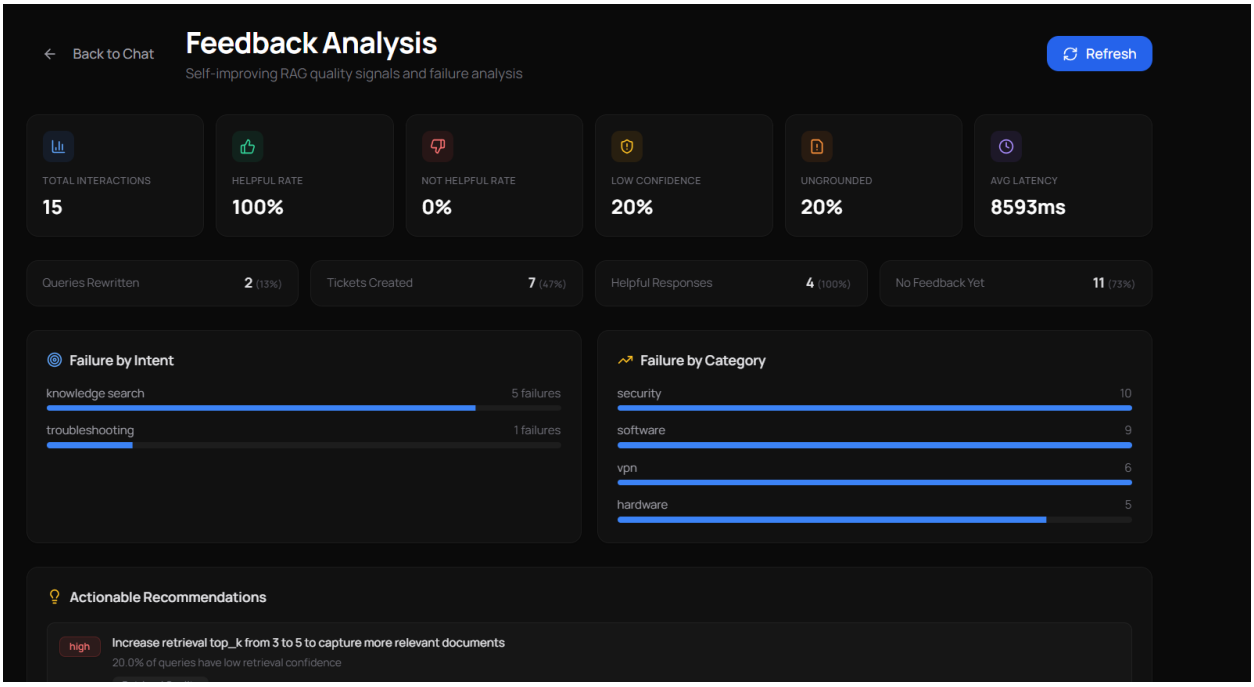
87.5 %

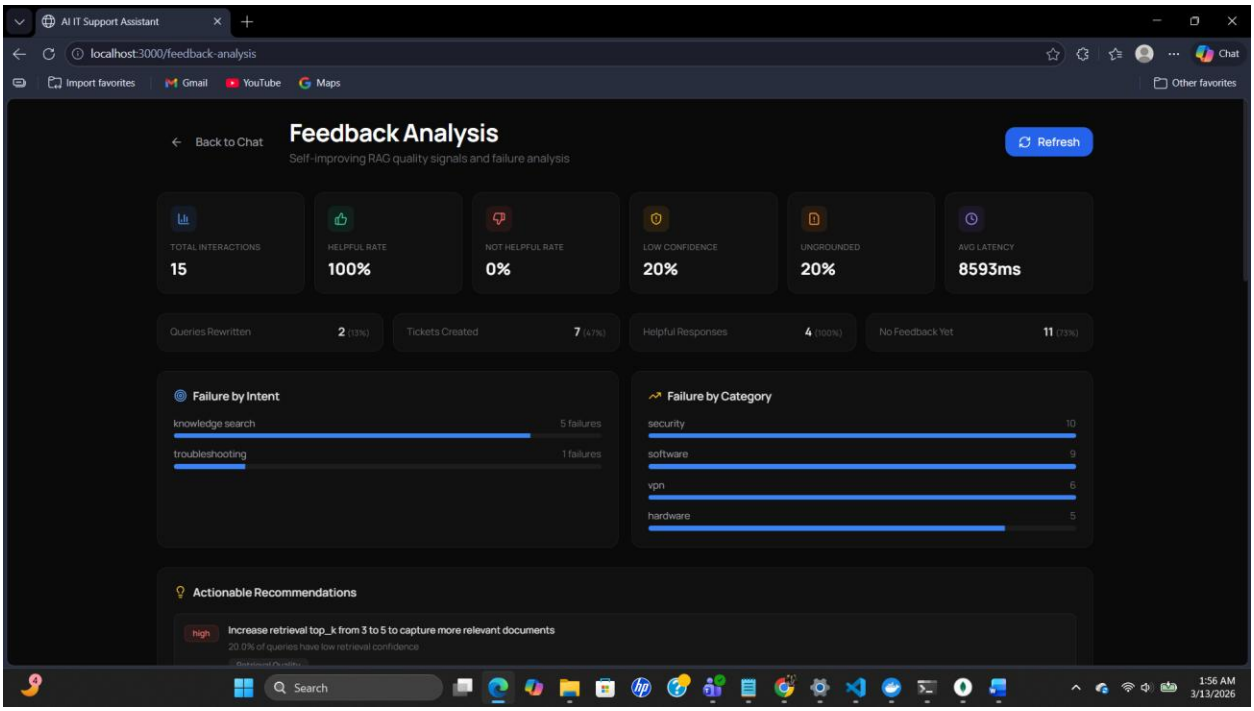
This tests the hybrid search engine on a newly added document. The query "My Laptop Wifi is not working" was matched against 12 knowledge base documents using two different search algorithms working together: TF-IDF cosine similarity (which understands the semantic meaning of "stuck" and "printing") and BM25 keyword matching (which specifically matches the word "Laptop"). The hybrid approach scored the new "Laptop and Wifi Troubleshooting" document (kb-009) as the top result. The AI generated detailed Laptop wifi troubleshooting. This proves the knowledge base can be expanded with new documents and they're immediately searchable.

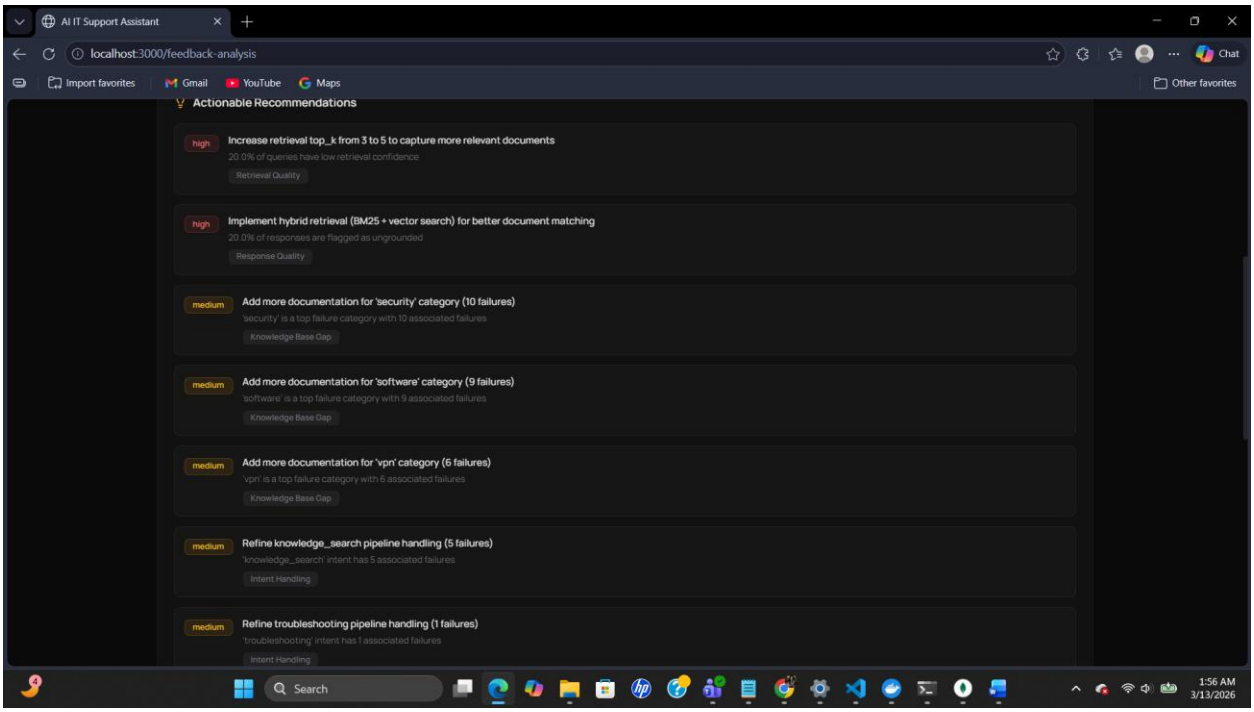



AI IT Support Assistant
localhost:3000/chat
Chat
Import favorites
Gmail
YouTube
Maps
Other favorites
IT Assistant
ENTERPRISE AI
New Conversation
What is the WiFi password for the office?
VPN Broken
My laptop wifi is not working.
How do I connect to the company VPN?
How do I reset my Okta MFA using the self
VPN broken
Create a ticket for laptop repair.
Evaluation Metrics
Feedback Analysis
Kumari Sabita 1311
Conversation
System Online
In Device Manager, right-click on the WiFi adapter and choose Update driver.
Select Search automatically for updated driver software.
e. Restart Your Laptop:
Reboot your laptop to ensure all settings and configurations take effect.
If Unresolved, recommend creating a support ticket:
If following these steps does not resolve the issue, please create a support ticket with your IT department detailing the steps you've taken and any error messages observed.
Action taken: A support ticket has been automatically created for follow-up. Ticket ID: 4529 | Priority: high | ETA: 24 hours
Helpful
Not Helpful
SOURCES
Laptop WiFi Troubleshooting
Laptop Hardware Issues and Repair
Slack Access and Configuration
Describe your IT issue or ask a question...
Search
2:15 AM
3/13/2026

This is the self-improvement intelligence dashboard. It aggregates all logged interaction data and surfaces actionable insights. The top row shows 6 key metrics: **Total Interactions** (how many queries processed), **Helpful Rate** (percentage of responses users rated positively), **Not Helpful Rate** (negative ratings), **Low Confidence** (percentage of queries where the search engine struggled), **Ungrounded** (percentage of responses that weren't well-supported by documents), and **Avg Latency** (average processing time). The second row shows operational stats: how many queries were rewritten, tickets created, and how many responses haven't received feedback yet. Below that, two breakdown charts show failure patterns: "Failure by Intent" reveals which query types are failing most, and "Failure by Category" shows which document categories need improvement. The "Actionable Recommendations" section provides prioritized suggestions like "Increase retrieval top_k from 3 to 5" or "Add more documentation for networking category."




AI IT Support Assistant
localhost:3000/feedback-analysis
Chat
Import favorites
Gmail
YouTube
Maps
Other favorites
Back to Chat
Feedback Analysis
Self-improving RAG quality signals and failure analysis
Refresh
TOTAL INTERACTIONS
15
HELPFUL RATE
100%
NOT HELPFUL RATE
0%
LOW CONFIDENCE
20%
UNGROUNDED
20%
AVG LATENCY
8593ms
Queries Rewritten 2 (13%)
Tickets Created 7 (47%)
Helpful Responses 4 (100%)
No Feedback Yet 11 (73%)
Failure by Intent
knowledge search 5 failures
troubleshooting 1 failures
Failure by Category
security 10
software 9
vpn 6
hardware 5
Actionable Recommendations
high
Increase retrieval top_k from 3 to 5 to capture more relevant documents
20.0% of queries have low retrieval confidence
Search
1:56 AM
3/13/2026


AI IT Support Assistant
localhost:3000/feedback-analysis
Chat
Import favorites
Gmail
YouTube
Maps
Other favorites
Actionable Recommendations
high
Increase retrieval top_k from 3 to 5 to capture more relevant documents
20.0% of queries have low retrieval confidence
Retrieval Quality
high
Implement hybrid retrieval (BM25 + vector search) for better document matching
20.0% of responses are flagged as ungrounded
Response Quality
medium
Add more documentation for 'security' category (10 failures)
'security' is a top failure category with 10 associated failures
Knowledge Base Gap
medium
Add more documentation for 'software' category (9 failures)
'software' is a top failure category with 9 associated failures
Knowledge Base Gap
medium
Add more documentation for 'vpn' category (6 failures)
'vpn' is a top failure category with 6 associated failures
Knowledge Base Gap
medium
Refine knowledge_search pipeline handling (5 failures)
'knowledge_search' intent has 5 associated failures
Intent Handling
medium
Refine troubleshooting pipeline handling (1 failures)
'troubleshooting' intent has 1 associated failures
Intent Handling
Search
1:56 AM
3/13/2026

AI IT Support Assistant

localhost:3000/feedback-analysis

Import favorites | Gmail | YouTube | Maps | Other favorites

medium **Optimize retrieval pipeline and consider caching frequent queries**

Average latency is 8593.0ms, above 5s threshold

Performance

low **Add a re-ranking stage after initial retrieval to improve document relevance**

Re-ranking consistently improves RAG precision in production systems

Advanced Retrieval

Sample Failure Cases

"How do I reset my Okta MFA?" medium conf.

Grounded: No (25%)

Reasoning: Keyword overlap: 3/12 content words found in source documents.

Add source documents that directly address this topic

"My VPN is not working and I'm blocked from accessing anything?" low conf.

Grounded: Yes (95%)

Reasoning: The response is well-grounded in the provided 'VPN Troubleshooting Guide'. Each troubleshooting step, such as checking the internet connection, restarting the VPN client, checking credentials, verifying system clock, and trying alternate gateways, is directly supported by the guide. The recommendation to create a support ticket if the issue persists is a reasonable extrapolation consistent with standard IT support practices, ensuring a high groundedness score.

Improve document coverage for this query category

Search

1:57 AM 3/13/2026

AI IT Support Assistant

localhost:3000/feedback-analysis

Import favorites | Gmail | YouTube | Maps | Other favorites

Re-ranking consistently improves RAG precision in production systems

Advanced Retrieval

Sample Failure Cases

"How do I reset my Okta MFA?" medium conf.

Grounded: No (25%)

Reasoning: Keyword overlap: 3/12 content words found in source documents.

Add source documents that directly address this topic

"My VPN is not working and I'm blocked from accessing anything?" low conf.

Grounded: Yes (95%)

Reasoning: The response is well-grounded in the provided 'VPN Troubleshooting Guide'. Each troubleshooting step, such as checking the internet connection, restarting the VPN client, checking credentials, verifying system clock, and trying alternate gateways, is directly supported by the guide. The recommendation to create a support ticket if the issue persists is a reasonable extrapolation consistent with standard IT support practices, ensuring a high groundedness score.

Improve document coverage for this query category

"What is the WiFi password for the office?" low conf. Helpful

Grounded: Yes (80%)

Reasoning: The response is supported by the notion that specific WiFi passwords are not typically included in general documentation, which aligns with the provided materials that detail troubleshooting and procedural guidance without mentioning specific passwords. However, it does not directly cite this procedure, as the source documents do not expressly state it, thus the response demonstrates some degree of inferred reasoning

Search

1:57 AM 3/13/2026